

```
In [1]: import numpy as np
```

```
In [4]: np.random.randint(10,40,(10,10)) # it generates 10*10 matrix (array) and from 10 to 39 (n-1)
```

```
Out[4]: array([[33, 23, 31, 18, 37, 34, 39, 12, 34, 23],
               [19, 34, 17, 25, 34, 25, 12, 20, 29, 27],
               [28, 21, 25, 15, 15, 32, 38, 34, 18, 34],
               [28, 20, 27, 12, 22, 28, 13, 28, 31, 23],
               [37, 35, 24, 13, 35, 11, 22, 36, 33, 33],
               [36, 27, 28, 10, 29, 16, 37, 21, 18, 36],
               [19, 19, 26, 39, 14, 27, 38, 19, 17, 32],
               [32, 38, 31, 32, 28, 27, 24, 30, 37, 35],
               [14, 13, 20, 28, 17, 39, 17, 32, 20, 29],
               [16, 20, 20, 31, 11, 20, 27, 20, 23, 29]], dtype=int32)
```

```
In [9]: np.random.randint(2,5,(5,5))
```

```
Out[9]: array([[3, 4, 2, 4, 3],
               [3, 2, 3, 2, 4],
               [3, 2, 2, 4, 2],
               [4, 3, 2, 2, 3],
               [2, 2, 3, 2, 4]], dtype=int32)
```

```
In [15]: b = np.random.randint(10,20,(5,4))
```

```
In [16]: b
```

```
Out[16]: array([[14, 18, 16, 10],
               [17, 18, 14, 18],
               [19, 16, 12, 19],
               [19, 18, 16, 17],
               [18, 18, 16, 17]], dtype=int32)
```

```
In [18]: type(b)
```

```
Out[18]: numpy.ndarray
```

```
In [19]: b[:] # : given in brackets displays complete array.
```

```
Out[19]: array([[14, 18, 16, 10],
               [17, 18, 14, 18],
               [19, 16, 12, 19],
               [19, 18, 16, 17],
               [18, 18, 16, 17]], dtype=int32)
```

```
In [20]: b[1:3]
```

```
Out[20]: array([[17, 18, 14, 18],
               [19, 16, 12, 19]], dtype=int32)
```

```
In [22]: b[1,2]
```

```
Out[22]: np.int32(14)
```

```
In [25]: b[1,3]
```

```
Out[25]: np.int32(18)
```

```
In [26]: b[2,3]
```

```
Out[26]: np.int32(19)
```

```
In [27]: b[1,-1]
```

```
Out[27]: np.int32(18)
```

```
In [29]: b[3,-2]
```

```
Out[29]: np.int32(16)
```

```
In [30]: b[2:3] # It displays complete 2 row .
```

```
Out[30]: array([[19, 16, 12, 19]], dtype=int32)
```

```
In [31]: b[0:-2]
```

```
Out[31]: array([[14, 18, 16, 10],  
               [17, 18, 14, 18],  
               [19, 16, 12, 19]], dtype=int32)
```

```
In [32]: b[-5,-3]
```

```
Out[32]: np.int32(18)
```

```
In [46]: b1 = np.random.randint(1,10)
```

```
In [47]: b1
```

```
Out[47]: 5
```

```
In [54]: b
```

```
Out[54]: array([[14, 18, 16, 10],  
               [17, 18, 14, 18],  
               [19, 16, 12, 19],  
               [19, 18, 16, 17],  
               [18, 18, 16, 17]], dtype=int32)
```

```
In [55]: b.reshape(3,2)
```

```
-----  
ValueError                                Traceback (most recent call last)
```

```
Cell In[55], line 1
```

```
----> 1 b.reshape(3,2)
```

```
ValueError: cannot reshape array of size 20 into shape (3,2)
```

```
In [70]: b1 = np.random.randint(2,10,(2,6))
```

```
In [71]: b1
```

```
Out[71]: array([[5, 2, 6, 5, 8, 4],  
               [6, 7, 2, 9, 5, 4]], dtype=int32)
```

```
In [72]: b1.reshape(3,4)
```

```
Out[72]: array([[5, 2, 6, 5],  
               [8, 4, 6, 7],  
               [2, 9, 5, 4]], dtype=int32)
```

```
In [101... import numpy as np  
import matplotlib.pyplot as plt
```

```
In [103... from PIL import Image
```

```
In [106... img = Image.open(r'C:\Users\user\Desktop\strw_img.jpg')
```

```
In [107... img
```

Out[107...



```
In [110... print(img)
print(type(img))
```

```
<PIL.JpegImagePlugin.JpegImageFile image mode=RGB size=304x166 at 0x25A904D57F0>
<class 'PIL.JpegImagePlugin.JpegImageFile'>
```

```
In [111... img = np.asarray(img)
img
```

```

Out[111... array([[ 50,  80,  10],
                [ 50,  80,  10],
                [ 50,  80,  10],
                ...,
                [ 42,  99,   5],
                [ 41,  98,   4],
                [ 41,  98,   4]],

               [[ 50,  80,  10],
                [ 50,  80,  10],
                [ 50,  80,  10],
                ...,
                [ 42,  99,   5],
                [ 41,  98,   4],
                [ 41,  98,   4]],

               [[ 50,  80,  10],
                [ 50,  80,  10],
                [ 50,  80,  10],
                ...,
                [ 43, 100,   6],
                [ 42,  99,   5],
                [ 41,  98,   4]],

               ...,

               [[212, 219, 229],
                [212, 219, 229],
                [211, 218, 228],
                ...,
                [201, 211, 220],
                [202, 212, 221],
                [205, 215, 224]],

               [[212, 219, 229],
                [212, 219, 229],
                [211, 218, 228],
                ...,
                [211, 221, 230],
                [213, 223, 232],
                [216, 226, 235]],

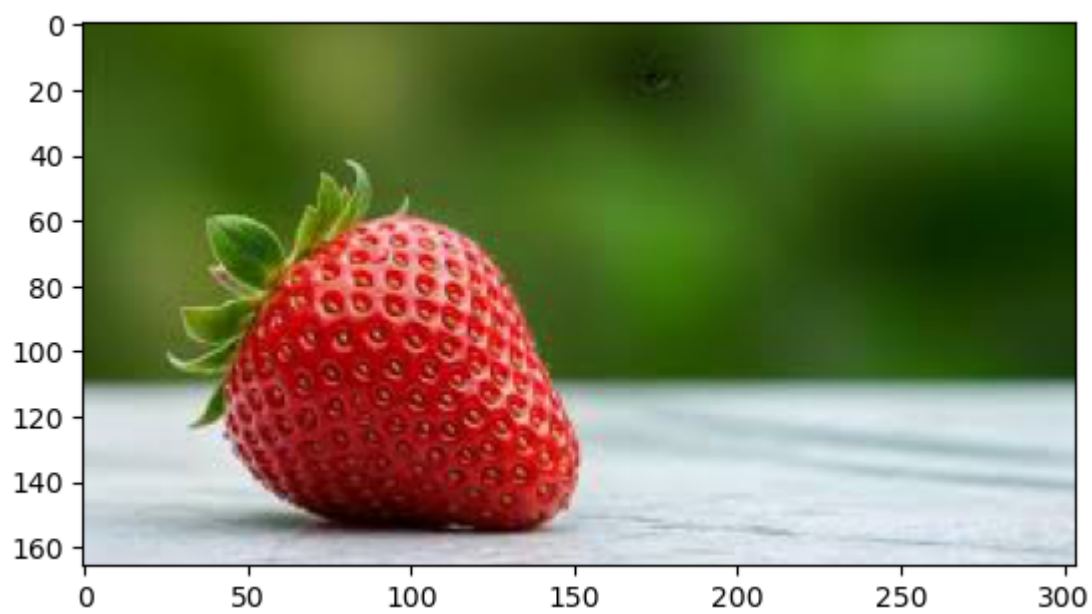
               [[212, 219, 229],
                [212, 219, 229],
                [211, 218, 228],
                ...,
                [209, 219, 228],
                [211, 221, 230],
                [213, 223, 232]]], shape=(166, 304, 3), dtype=uint8)

```

```

In [112... plt.imshow(img)
            plt.show()

```



In [114...] `plt.imshow`

Out[114...] `<function matplotlib.pyplot.imshow(X: 'ArrayLike | PIL.Image.Image', cmap: 'str | Colormap | None' = None, norm: 'str | Normalize | None' = None, *, aspect: "Literal['equal', 'auto'] | float | None" = None, interpolation: 'str | None' = None, alpha: 'float | ArrayLike | None' = None, vmin: 'float | None' = None, vmax: 'float | None' = None, colorizer: 'Colorizer | None' = None, origin: "Literal['upper', 'lower'] | None" = None, extent: 'tuple[float, float, float, float] | None' = None, interpolation_stage: "Literal['data', 'rgba', 'auto'] | None" = None, filternorm: 'bool' = True, filterrad: 'float' = 4.0, resample: 'bool | None' = None, url: 'str | None' = None, data=None, **kwargs) -> 'AxesImage'>`

In [113...] `img.shape`

Out[113...] `(166, 304, 3)`